

Sample Quiz 1 solutions

DATA 412/612

1. subset()

```
offers <- data.frame(
  firm   = c("Helix", "Nimble", "Helix", "Orbit", "Nimble", "Orbit", "Helix"),
  role   = c("intern", "intern", "analyst", "intern", "analyst", "analyst", "intern"),
  salary = c(22, 19, NA, 21, 28, 27, 24),
  remote = c(TRUE, FALSE, TRUE, TRUE, FALSE, TRUE, TRUE)
)
```

(a)

Intern salaries: 22, 19, 21, 24. Mean with `na.rm = TRUE` is **21.5**. Keep rows with salary 22 and 24, then drop role.

The mean must be computed **after** the intern rows are isolated. One `subset()` that writes

```
role == "intern" & salary > mean(salary, na.rm = TRUE)
```

uses the mean of **every** salary (23.5), so it keeps only the 24. That is wrong.

```
offers |>
  subset(role == "intern") |>
  subset(salary > mean(salary, na.rm = TRUE), select = -role)
```

	firm	salary	remote
1	Helix	22	TRUE
7	Helix	24	TRUE

Equivalent without a pipe:

```
interns <- subset(offers, role == "intern")
subset(interns, salary > mean(salary, na.rm = TRUE), select = -role)
```

(b)

`subset()` on a vector drops values whose condition is NA. Indexing does not:

```
subset(offers$salary, offers$salary >= 22)
# [1] 22 28 27 24

offers$salary[offers$salary >= 22]
# [1] 22 NA 28 27 24
```

Correct answer: 22 28 27 24.

2. pair_mean_call

```
pair_mean_call <- function(df) {
  n <- ncol(df)
  if (n < 2) {
    return("thin")
  }
  if (n > 2) {
    return("wide")
  }

  a <- df[[1]]
  b <- df[[2]]
  if (!is.numeric(a) || !is.numeric(b)) {
    return(FALSE)
  }
  if (any(is.na(a)) || any(is.na(b))) {
    return("missing")
  }

  m1 <- a |>
    subset(is.finite(a)) |>
    mean()
  m2 <- b |>
    subset(is.finite(b)) |>
    mean()

  if (m1 >= m2) {
    return(1)
  }
  names(df)[2]
}
```

Input	Result	Why students miss it
<code>data.frame(u = 1:2)</code>	"thin"	<code>nrow()</code> would be the wrong test
<code>data.frame(u = 1:2, v = c("a", "b"))</code>	FALSE	<code>is.numeric(df)</code> is always FALSE; test the columns
<code>data.frame(u = c(1, NA), v = c(2, 3))</code>	"missing"	<code>if (is.na(a))</code> is not a single TRUE/FALSE
<code>data.frame(u = c(1, Inf, 3), v = c(10, 10, 10))</code>	"v"	Inf is not NA. <code>mean(c(1, Inf, 3))</code> is Inf, so <code>Inf >= 10</code> wrongly returns 1. Finite mean of u is 2
<code>data.frame(low = c(2, 4), high = c(1, 1))</code>	1	first mean $3 \geq$ second mean 1
<code>data.frame(low = c(1, 1), high = c(2, 4))</code>	"high"	return <code>names(df)[2]</code> , not the literal "second"

Suggested checks:

```
pair_mean_call(data.frame(u = 1:2))
pair_mean_call(data.frame(u = 1:2, v = c("a", "b")))
pair_mean_call(data.frame(u = c(1, NA), v = c(2, 3)))
```

```
pair_mean_call(data.frame(u = c(1, Inf, 3), v = c(10, 10, 10)))  
pair_mean_call(data.frame(low = c(2, 4), high = c(1, 1)))  
pair_mean_call(data.frame(low = c(1, 1), high = c(2, 4)))
```